Developing an XAI-Based Crop Recommendation Framework Using Soil Nutrient Profiles and Historical Crop Yields

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***Abstract*—The agriculture sector is a significant cornerstone of the country and has boomed drastically in past decade, due to the advent of cutting-edge technologies such as Artificial Intelligence (AI), Machine Learning (ML), Blockchain, Cloud Computing, and Internet of Things (IoT). Recent studies have utilized ML models to overcome challenges like variability in climate patterns, soil fertility depletion, water scarcity, market volatility and optimal crop recommendation, by considering the features such as soil nutrients, climate conditions, rainfall, temperature, and fertilizers that significantly transform the agricultural sector. However, transparency, interpretability, and accountability of underlying models' decisions are often neglected in real-life use cases. Hence, the authors have proposed a hyperparameter optimization-based grid search algorithm for smart agriculture framework that recommends the optimum crop with high accuracy utilizing the vital features. The proposed framework offers clear explanations, and transparency for each crop recommended to the farmers and demonstrates the key factors influencing the decision-making process using eXplainable-AI (XAI). Furthermore, the study is compared with conventional ML models such as Logistic Regression (LR), Decision Tree Classifier (DTC), Random Forest Classifier (RFC), K-Nearest Neighbor (KNN), Support Vector Machine (SVM), and Gradient Boosting (GB) to measure the performance. Consequently, the hyperparameter optimization-based grid search algorithm outperforms other state-of-the-art approaches in terms of accuracy up to 99.73%, precision by 95.05%, RMSE value by 99.52%, MAE value by 99.36% and R2 value by 99.17 thereby enhancing the validation and transparency of crop predictions by potential integration of XAI-based approaches.**

***Index Terms*—Crop recommendation, hyperparameter-optimization, explainable-AI, machine learning, smart farming.**

I. Introduction

AGRICULTURE is undeniably the most imperative and predominant sector, serving as the backbone for feeding the ever-growing demands of an expanding global population. Apart from this, it is the largest source of livelihood for farmers, with a significant portion of the population, depending on farming for their survival. Hence, the agricultural sector provides crucial employment opportunities, supporting the livelihoods of millions of people around the world. In 2022, over half of the workforce in India worked in farming, adding between 17% -18% to the national land area, which remained consistent at around 328.7 million hectares from 1971 to 2020. By 2025, the Indian agriculture sector growth 24 USD billion, and 70% of its revenue generated through retail sales, positioning it as the sixth-largest market for food and groceries globally. According to the survey, the last financial year 2023-2024 indicates that the agriculture sector produces a total of 328.8 million tons of grains. In the upcoming year 2024-25 Indian government set the production target at 340.40 million tons [1]. The highest crop production occurs during the kharif season, while the lowest is during winter. Farmers have the potential to increase production in favorable conditions and reduce production loss in unfavorable conditions through the use of crop yield estimation. Different factors, like agricultural methods, decisions, the use of pesticides and fertilizers, climate conditions, and the cost of selling products, can affect optimistic predictions regarding the amount of crops harvested.

However, traditional agricultural practices are no longer sufficient to sustain the growing population while simultaneously conserving nature's limited resources. As a result, farmers across the globe are facing several challenges such as unpredictable climate conditions, water scarcity, soil fertility depletion, excessive and indiscriminate deployment of chemical fertilizers, and pesticides, market volatility, and many more. Therefore, to address these gaps, smart agriculture or precision agriculture emerged which aims at enhancing cutting-edge technologies and data analytics toward smart decision-making. Smart-driven farming comprises various tools such as IoT sensors, drones, and Artificial Intelligence (AI) techniques to offer solutions to optimize farming practices, increase productivity along prioritize sustainability [2].

The ML model of the crop recommendation system (CRS) leverages soil nutrient and crop yield data to provide agronomist with data-driven suggestions for optimal crop selection [2]. The machine learning (ML) model helps improve farming productivity, reduce resource wastages, and promote sustainable farming practices by aligning crop prediction with the inherent characteristics of the soil. The ML-based framework often relies heavily on the data they are trained on, which may represent specific regions, types of soil, climate conditions, and farming practices. If applied to areas with significantly different environmental conditions, the model recommendations might be suboptimal or inaccurate. The ML models might struggle with dynamic variables like water availability, extreme weather conditions, or sudden changes in market conditions that are not represented in the dataset.

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The authors have incorporated a hyperparameter optimization-based grid search algorithm with the framework for optimal crop recommendation based on soil nutrients and historical crop yield to enhance overall accuracy and robustness. This study reduces the risk of overfitting, as aggregates predictions and effectively balances the strength of different algorithms. Hyperparameter optimization-based grid search algorithm improves reliability by providing more stable and consistent crop recommendations. Integrating eXplainable AI (XAI) in a crop recommendation system provides significant benefits by enhancing model transparency, trustworthiness, and usability [3], [5]. XAI models make the decision-making procedure of the AI system interpretable, allowing farmers and agronomic experts to understand how specific crop recommendations are generated.

Motivation

In recent years, farmers have faced various challenges such as declining soil fertility, inefficient use of resources, and unpredictable crop yield due to varying climate conditions and nutrient imbalances. The Machine Learning (ML) models were employed to uncover complex patterns and interactions between soil nutrients, climate conditions, and crop performance, but failed to resolve the issue completely. The hyperparameter optimization-based grid search approach enhances reliability by providing more stable and consistent crop recommendations. However, the adoption of crop recommendation systems in smart agriculture is often impeded by the lack of clarity in AI models. These situations the urgent need for XAI-based models capable of providing clear, transparent, and interpretable insight to farmers. By integrating XAI techniques, the framework aims to analyze the relationship between soil properties, historical performance, and environmental factors to provide data-driven and precise crop recommendations. XAI improves decision-making and supports sustainable farming tailored to specific soil and climate conditions. This study presents various important factors of XAI-based crop predictions, highlighting XAI's importance and its applications to revolutionize smart agricultural solutions. Special contribution to the article is given below:

- To propose a smart agriculture framework integrated with hyperparameter optimization based grid search model for a crop recommendation system based on soil nutrient and historical crop yield combining the predictive power of multiple base models to enhance accuracy and performance.
- The constructed framework is powered by XAI and enhances transparency, farming trust, and usability with high accuracy. It's adaptable to diverse agronomic conditions and can scale across regions, providing versatile tools for global farming challenges.
- The experimental results indicate superior performance of the proposed XAI-enabled hyperparameter optimization-based grid search algorithm integrate into an agriculture framework with improved influential parameters.
- These farming contributions can be useful for practical farming applications, theoretical improvement, and broader impacts on the agronomist community.

The consequence of the article goes as follows: Section II explores the various related work to this study. Sections III present the proposed various ML models for crop recommendation systems define the experiment setup of the proposed work and explain the setup tools and library. Section IV presents result and discussion. Section V defines the conclusion and future scope works.

II. Related Work

The authors depict crop classification approach and crop yield prediction using ML techniques for the agriculture sector [3]. It also recognizes critical limitations related to data quality, agriculture infrastructure, model interpretability, and the need for ongoing research to enhance the applicability of its findings. The authors have proposed a crop recommendation system for Agriculture 5.0 presents a sophisticated method for enhancing agriculture productivity [4]. This system integrates ML techniques and IoT, to provide real-time, data-driven insight for agronomists. Khan et al. [3] proposed an IoT-based fertilizer recommendation, that has shown significant potential in enhancing crop productivity and sustainability. Future work is needed to integrate advanced ML algorithms for system refinement. Nikam et al. [4] have deployed an XAI-based LIME algorithm for species identification to address the opacity and complexity of the traditional XAI model which obstructs trust and adoption among end-users. Another work by Shams et al. [5] introduced crop recommendation with XAI. The main objectives find the gap between XAI-driven automation and farmer understanding empower with actionable insights and facilitate trust in XAI-based crop recommendations.

The authors have applied XAI techniques to crop recommendation systems to enhance transparency and interpretability. LIME and SHAP are among the most commonly used methods for providing post-hoc explanation of AI models' predictions. Sani et al. [6] proposed AI-based techniques for smart data-driven crop prediction to address the challenge of data scarcity and heterogeneity in agricultural settings, the authors have explored transfer learning and fusion models to improve the generalization and robustness of crop recommendation models. Sagar et al. [7] used a crop recommendation system combining information from diverse sources such as remote sensing data, weather forecasts, and soil quality measurements, the proposed approach enables specific and context-aware recommendations tailored to specific crop conditions. Elavarasan and Vincent [8] Discuss a crop prediction using reinforcement learning. They employed Q-Reinforcement learning to interpret the model's decisions, providing insights into how various features (e.g., soil type, pH, weather conditions) influenced the recommendations. Li et al. [9] focused on predicting weed density detection methods while ensuring the PSP net model.

By generating crop segmentation and using the color index for individual predictions, the system highlighted the most significant features affecting crop yield predictions, such as rainfall and temperature. Mary and Kumar [10] The XAI model has been integrated into precision agriculture frameworks to enhance farming practices and crop predictions.

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TABLE I
STATE-OF-THE-ART COMPARISON TABLE FOR EXPLAINABLE AI IN SMART AGRICULTURE

Author	Approach	Objective	Advantages	Limitations
Gokila et. al [13]	Rule-based System	Provide Clear, rule-based recommendation	Easy to understand and Implementation	Limited by the quality and completeness of predefined rules
A.Reyana et. al [14]	Decision Tree, Random Forest (RF), Hoeffding Tree Algo.	Offer transparent decision-making paths.	High interpretability and easy-to-visualize	Can become complex with too many features; prone to overfitting
Arfat et.al [3]	SVM, LR, GNB, KNN	Identify key factors influencing recommendations	Provides insight into feature importance	Aggregate models are less interpretable than individual trees
Melvin et. al [15]	SHAP (Shapley Additive exPlanation)	Explain individual predictions by assigning feature importance	Offers consistent and locally accurate explanation	Computationally intensive for large datasets
Kiran et. al [16]	LIME Algorithm	Explain individual predictions with locally approximated models	Model-agnostic; can be applied to any AI model	May produce different explanation for similar instances
Dalhatu et. al [17]	Clustering based recommendation	Recommendations based on similar historical cases	High transparency and relevance	Requires a comprehensive and representative case database
Jamalbek et. al [18]	Vanilla CNN, LR, XGBoost	Provide interpretable predictions from complex models	The balance between performance and interpretability	May still require domain-specific knowledge to fully understand
Padmavathi et. al [19]	Attention Mechanisms in Deep Learning	Highlight relevant parts of input data in predictions	Enhances interpretability of deep learning models	Interpretation depends on the context and may not always be clear

These proposition models leverage decision tree (DT) and gradient boosting classifier (GBS), enriched with crop feature importance analysis and SHAP values to provide global interpretable insight into model decisions. Alzubi and Galyna [11] Used a sustainable agriculture model combining XAI with a deep learning framework for crop recommendations. By deploying a deep neural network for predicting crop analysis and XAI-based SHAP algorithm for interpretability, the system offered recommendations aligned with sustainability crop objectives, and analysis of different crop parameters such as soil properties, climate conditions, and crop nutrients. Das and Chatterjee [12] They have used an XAI-based machine learning model to recommend crops using IoT-based sensor data. XAI-based SHAP model was employed to interpret model outputs, highlighting the influence of climate variables like rainfall, temperature, and crop selections. Several researchers have applied ML-based models or XAI-based algorithms to find the best crop for the farmer in a particular area with transparency and high accuracy, but most of the proposed approaches face the challenges of overfitting, transparency, trustworthiness, and computational complexity as given in Table I [16], [17], [18], [19], [20], [21], [22].

Nurcahyo et al. [20] Proposed XAI-based multi-class crop management system. This article used machine learning techniques to generate crop prediction, while the SHAP algorithm explained the impact of climate conditions and historical crop data on the recommendations provided. Turgut et al. [21] Developed an edge computing-based XAI crop recommendation system, the model addresses the challenges posed by climate change and resource limitations by providing tailored crop suggestions based on the local environment and soil conditions. Mohan et al. [22] Aimed to design AI and XAI-based models for precision crop yield predictions. The developed model focuses on temperature, rainfall, and macronutrient level may overlook other critical variables that could impact crop yield predictions. Soni et al. [23] Discuss the enhanced neural network designed for predicting soil moisture in precision agriculture. This advancement boosts agricultural efficiency by refining irrigation schedules, maximizing the use of resources, and reducing water waste. Bwambale et al. [24] Discuss the evaluation of model predictive control (MPC) within the context of precision agriculture. This includes a summary of the fundamental principles of MPC, its formulation, and the optimization methods employed, highlighting its predictive capabilities and adaptability. Diachenko et al. [25] The author examines a decision support system designed to predict the risk of wheat powdery mildew by utilizing weather monitoring and an explainable artificial intelligence (XAI) algorithm. Rubo and Zinkernagel [26] Explore the improvement of irrigation demand forecasting for open field vegetable crops in Germany by utilizing neural networks, transfer learning, and ensemble learning models. Abekoon et al. [27] Discuss explainable AI to predict soil nutrient content in cabbage cultivation, emphasizing experiment were conducted on cabbage plants growth in central hills. This study provides reliable, interpretable, and sustainable crop prediction, this system empowers farmers with data-driven insight, enabling more efficient and trustworthy smart farming practices.

III. XAI ENABLED PROPOSED FRAMEWORK FOR SMART AGRICULTURE

The proposed framework focuses on Indian agriculture sector due to its varying climatic conditions, ranging from humidity and dry conditions in the south to temperate alpine conditions in the north of India. India's land border

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is 328.7 million hectares, extending from the snow-capped Himalayan peaks to the tropical rainforests in the southern central region. Indian agriculture land, 297.3 million hectares were used for agriculture between 1971 and 2020. The smart agriculture system utilizing soil nutrients and historical crop yield often relies on ML algorithms to analyze complex agriculture data and, the non-linear relationship between soil properties and crop productivity. We have used a crop recommendation dataset that is downloaded from Kaggle. This dataset was created by enhancing augmenting datasets that include pH, rainfall, climate, and fertilizer information pertinent to India. First, we read the dataset and then preprocessing data. During preprocessing, we handled the missing data and applied feature engineering techniques. We have applied data normalization techniques to standardize categorical variables such as crop types using one hot encoding technique and complete the data normalization process, applying the Z-score analysis techniques to detect values in soil nutrient levels. After the data normalization process, compute the Pearson correlation coefficients analysis metrics to identify highly correlated crop features (eg. N, P, K, Rainfall, humidity) that may introduce variable redundancy. K-fold cross-validation methods enhance the credibility of the findings and provide a more comprehensive evaluation of the framework performance in real-world agriculture sectors. We have used the 5-fold cross-validation method and split the dataset into 5 equal parts (folds) using 4 folds for training purposes and 1 fold for testing purposes in each iteration. We have evaluated our model performance metrics such as accuracy, precision, recall, F-1 score, and RSME for each fold.

The data has been divided into three parts: the train set, the test set, and the validation set. 70% of the whole data is chosen for model training, while the rest, 30%, is reserved for model testing and validation [28]. This study integrated various machine learning models such as (LR, DT, KNN, RF, SVM, Gradient Boosting, and Grid Search) models and calculates the model performance and evaluations. We have used a Google Cloud server to store the results of the crop recommendation system on the servers and structured data, including recommended crop, soil parameters and yield predictions stored in the cloud database for easy retrieval and data analysis. Integrating a cloud-based machine learning pipeline with automated backups ensures real-time updates and integration with farming applications. The proposed model enables remote access, scalability, and efficient management of agriculture data, supporting informed decision-making for farmers and agronomists.

When calculating the model performance and then integrate XAI techniques that leverage soil nutrient and crop yield predictions by increasing model transparency, trust, and decision making. XAI-enabled Crop Recommendation Framework leverages advanced ML techniques to provide transparent and interpretable crop selection guidance for farmers. LIME seeks to function as the 'explainer', providing clarifications for the predictions associated with each data sample [29]. These values assist in identifying biases and errors in the model. By examining feature contributions, users can detect unexpected patterns and enhance the model's accuracy and fairness.

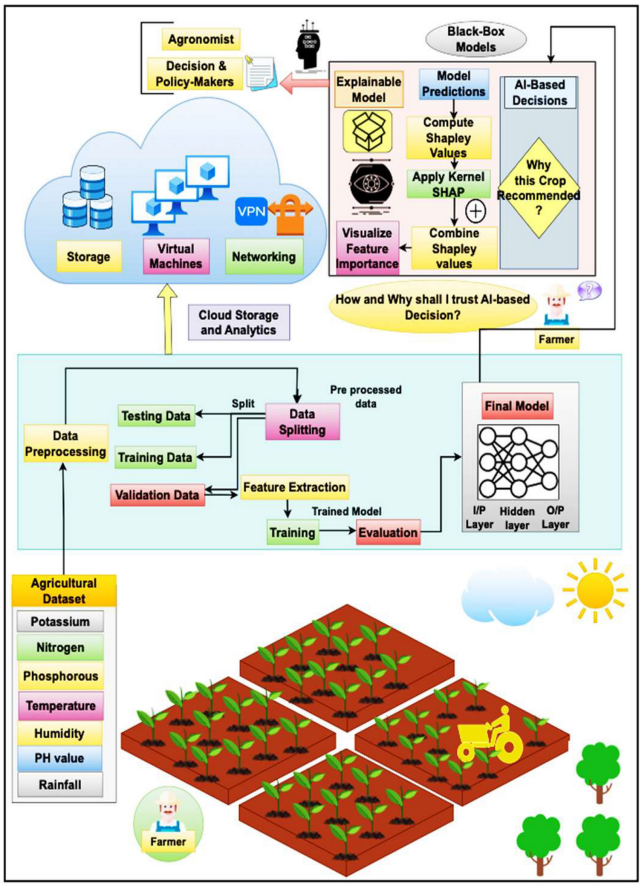

Fig. 1. Proposed XAI-enabled model for CRS.

Figure 1 employs a crop recommendation system with XAI techniques, the framework explains the major crop factors influencing its recommendations, and ensuring farmers can trust and understand the smart farming techniques. Model transparency not only enhances farmers' confidence in adopting AI-driven recommendations but also allows for fine-tuning and validation of the model with local agronomic expertise, ultimately promoting sustainable and optimized agricultural practices [30]. Grid Search-based XAI model helps farmers understand which soil properties impact crop selections. For example, farmers in a region with high nitrogen (N), low potassium (K), and moderate pH want to know which crop is planted, the traditional ML model recommends "Maize" with no explanation. XAI model recommends "Maize" with LIME value explanation such as +0.40 from nitrogen (N), -0.25 from potassium (K), and +0.18 from pH values. The farmers now understand the reasoning and can adjust soil conditions to improve crop yield. Hence, XAI explains the internal process of a model and it's used to explain the method, procedure, and output of the model that should be understandable by the users. The main aim is to recommend better crop growth based on different crop parameters while ensuring the model's decisions are interpretable and transparent.

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IV. Proposed Hyperparameter Optimization-Based Grid Search Algorithm and State-of-the-Art Implementation

This study constructed a hyperparameter optimization-based grid search algorithm for crop recommendation using soil nutrients and historical crop yield, with the grid search algorithm's most efficient method due to its ability to learn from previous evaluations and select the next parameters combination to test the model performance. It plays an important role in improving the performance of a deep learning-based crop recommendation system. The main objective of the grid search algorithm is to identify the best set of hyperparameters that maximize model performance while ensuring the model is interpretable and robust across different soil conditions. It can be used to fine-tune parameters such as learning rate, batch size, dropout rate, number of layers, and feature selection to enhance the predictive accuracy of the crop recommendation system. This model addresses the challenges of traditional crop recommendation systems by offering robust crop prediction; enhanced transparency, and actionable insight for improving soil health and crop productivity.

Algorithm 1 Hyperparameter Optimization-Based Grid Search Algorithm

Input: C: Set of possible crop
S: represents soil nutrient profile
H: Historical crop
Λ: Hyperparameter for Grid Search
T: Decision threshold
Output: Recommend crop $C^*$with highest suitability score

Read soil nutrient profile S and historical yield data H
Normalize data to bring soil properties
$x_i = x_i - \min(x)/\ max(x) - min(x)$
Divide the dataset into train test split data
Model training with Grid search
Define hyperparameter_grid Λ
$\lambda \in \Lambda$
$f(x,\ \lambda)$
$\lambda^* = \underset{\lambda}{\arg\min}\ L(y, f(x,\ \Lambda))$
$\widehat{y}_c = f(x,\ \Lambda^*)$
$\varphi_i = \mathbb{E}\ [\ f(x)|x_i] - \mathbb{E}[f(x)]$
$Importance\ (s_i) = \sum_{j=1}^{n} \left| \varphi_{i,j} \right|$
$CSI_c = \sum_{i=1}^{m} w_i.s_i + \widehat{y}_c$
$c^* = \underset{c \epsilon C}{arg\ max}\ CSI_c\ if\ CSI_c \geq T$
Evaluate model performance metrics

A. Hyperparameter Optimization-based Grid Search Model

ML model is used to predict the crop or suitability score for different crops. Let C be the set of possible crops $C = \{c_1,\ c_2, \ldots.., c_n\}$ and S represent soil nutrient features such as $S = \{s_1,\ s_2, \ldots..,\ s_n\}$ and the H represent the historical crop yield for a corresponding crop such as $H = \{h_1,\ h_2, \ldots.., h_n\}$

$$X \epsilon\ \mathbb{R}^{mXn}; \tag{1}$$

These represent the soil nutrient matrix for different samples $Y \epsilon\ \mathbb{R}^n$; Representing the historical yield for different crops.

$$C^* = \underset{c \epsilon C}{arg\ max}\ \mathbb{E}\big[f\ (x,\ \theta_c)\big] \tag{2}$$

where $f\ (x,\ \theta_c)$ represents predictive functions that estimate the yield potential of crop c given soil feature x.

θ_c: denoted the hyperparameters for the crop.

$\mathbb{E}$: Represent the expected crop yield.

An ML Model can be trained using the soil profiles and historical yields:

$$\hat{y} = f(x, \theta_c) \tag{3}$$

θ: Represent the model hyperparameters.

Let λ represent the set of possible hyperparameters chosen by minimizing the loss function.

$$\lambda^* = \underset{\lambda}{\arg\min} \sum_{i=1}^{n} L(y_i,\ f(x_i;\ \lambda)) \tag{4}$$

where $L(y_i,\ \widehat{y}_i)$ defines the loss function.

$$L\big(y_i, \hat{y}_i\big) = \frac{1}{n} \sum_{i=1}^{n} \big(y_i,\ \hat{y}_i\big)^2 \tag{5}$$

To enhance model transparency and interpretability XAI algorithms such as LIME and SHAP are used to analyze feature contributions.

$$\varphi_i = \mathbb{E}\big[f(x)|x_i\big] - \mathbb{E}[f(x)] \tag{6}$$

φ_i: represent the SHAP values and define the importance of feature s_i.

$\mathbb{E}[f(x)|x_i]$: is the expected conditioned-on feature s_i

$\mathbb{E}[f(x)]$: Define the expected prediction overall feature combinations.

Feature importance score:

$$Importance\ (s_i) = \sum_{j=1}^{n} |\varphi_{i,j}| \tag{7}$$

Define the crop suitability index (CSI) based on the weight factor of soil features and historical yields.

$$CSI_c = w1.N + w2.P + w3.K + w4.pH + \ldots \tag{8}$$

where w_i represents the importance weight obtained from the model.

$$CSI_c = \sum_{i=1}^{m} w_i.s_i + \hat{y}_c \tag{9}$$

The recommended crop is:

$$C^* = \underset{c \epsilon C}{arg\ max}\ CSI_c \tag{10}$$

The final crop recommendation is based on a threshold T

$$Recommend\ c^* if\ CSI_c \geq T \tag{11}$$

where T is a predefined acceptance threshold.

Proposed a hyperparameter optimization algorithm for a crop recommendation system

Where: c^* represents the highest suitability score and Λ defines the hyperparameter grid that includes the learning rate, number of estimators, and regularization parameter.

B. Logistic Regression

We employ logistic regression, a statistical technique for classifying data into two categories, to consider past records

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on agricultural production, characteristics of the soil, climate conditions, and additional pertinent factors.

Input features such as soil pH, rainfall, temperature, soil type, etc.

$$X_1,\ X_2,\ X_3 \ldots\ldots\ldots\ldots X_n \tag{12}$$

K: Target variable representing the crop type.

$$P(K = 1|X)$$

$$P(K = 1|X) = \sigma(\beta_0 + \beta_1 X_1 + \beta_2 X_2 \ldots\ldots\ldots\ldots + \beta_n X_n) \tag{13}$$

where σ is the sigmoid function defined as:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{14}$$

$\beta_0,\ \beta_{1,\ \beta_2}$: Coefficients of the model learned from the training data.

$$L(\beta) = -\frac{1}{m}\sum_{j=1}^{m}\left[K_i \log\left(\hat{K}_i\right) + (1 - K_i)\log\left(1 - \hat{K}_i\right)\right] \tag{15}$$

C. Decision Tree

We deploy a decision trees classifier performing multi-class classification on a dataset. We have analyzed crop recommendation data that included optimal conditions for different crops and generated clear, rule-based recommendations [25]. The tree consists of internal nodes representing decision points on features, branches representing the output of those decisions, and leaf nodes representing the final classification or output.

Gini impurity can be used to determine the best split at each node, such as:

$$Gini(D) = 1 - \sum_{k=1}^{K} P_k^2 \tag{16}$$

Here P_k defines the probability of class k.

$$IG(D,\ A) = H\ (D)\ -\ \sum_{v \epsilon Values(A)} \frac{|D_v|}{|D|}\ H(D_v) \tag{17}$$

where H (D) is the entropy of the dataset D, and D_v is the subset of D where feature A takes the values v.

$$H(D) = -\sum_{k=1}^{K} P_k \log_2\ (P_k) \tag{18}$$

Gain Ratio $(D,\ A) = \frac{IG(D,\ A)}{H_A(D)}$: Where $H_A(D)$ is the intrinsic value of the feature A.

$$H_A(D) = -\sum_{v \in \text{Values}(A)} \frac{|D_v|}{|D|}\ \log_2\ \frac{|D_v|}{|D|} \tag{19}$$

D. Random Forest

We deploy a random forest algorithm to build a variety of decision models during the training phase and return the most common class (classification) or average prediction (regression) of each model. By analyzing comprehensive datasets that include soil properties, weather conditions, and historical crop performance, the random forest algorithm identifies patterns and relationships that influence crop success. For a dataset D with m samples, generate B bootstrap samples $D_1,\ D_2 \ldots\ldots\ldots D_B$ by sampling with replacement.

The Gini impurity for a node t is defined as: $(t) = 1 - \sum_{k=1}^{K} P_k^2$; where P_k is the proportion of class k instance node t.

$$IG(D,\ A) = H\ (D)\ -\ \sum_{v \epsilon Values(A)} \frac{|D_v|}{|D|}\ H(D_v) \tag{20}$$

H (D) defined the entropy of the dataset D:

$$H(D) = -\sum_{k=1}^{K} P_k \log_2\ (P_k) \tag{21}$$

For each node, randomly select F features from the total n features. Choose the best feature A and split the point based on the criterion.

$$\widehat{Y} = \arg\ \max_{\text{yk}}\ \sum_{i=1}^{B} \delta\left(\text{y}_i' = \text{yk}\right) \tag{22}$$

where $\delta(\cdot)$ defines the indicator function. If the argument is true then the function returns1 otherwise returns 0.

E. K-NN Classifier

KNN classifier is a ML technique used for classification that classifies crops based on the crop similarity of new data points to existing data points. By analyzing key factors such as crop parameters as we discussed earlier, the KNN algorithm identifies the most similar instances in the dataset and recommends crops that have performed well under similar conditions [23]. To find the nearest neighbor, we need a distance metric to calculate the similarities of crops.

Euclidean distance matrices:

$$d\left(X,\ X'\right) = \sqrt{\sum_{i=1}^{n}\left(X_i - X_i'\right)^2} \tag{23}$$

where X and X′ are two points in the feature space: where X is the new data point, the KNN classifier finds the K training set examples that are closest to X based on the chosen distance metric:

$$\text{Neighbor}(X)\ = X_1',\ X_2', \ldots\ldots\ldots X_k' \tag{24}$$

$$\hat{Y} = \arg\ \max_{\text{yk}}\ \sum_{i=1}^{k} \delta\left(y_i' = yk\right) \tag{25}$$

Given $\delta(\cdot)$ indicate function values that return 1 if the argument is true and the argument is false return 0, and y_i' define the class label.

F. Support Vector Machine

Our model leverages SVM to classify and predict crop suitability based on crop recommendation dataset. The proposed model calculates the optimal hyperplane that differentiates crop outcomes.

For binary classification, the decision function of a linear SVM is:

$$f(X) = \beta_0,\ +\beta_1 X_1 +\ \beta_2 X_2 \ldots\ldots \beta_n X_n \tag{26}$$

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The objective is to find the weight vector $\beta = \beta_0,\ \beta_1,\ \beta_2 \ldots\ldots \beta_n$ and bias β_0 that maximum the margin while minimum classification errors.

$\{(X_i,\ Y_i)\}_{i=1}^{m}$: Where X_i are the input features and Y_i are the class labels.

The optimization problem for a hard margin SVM is:

$$min_{\beta,\beta_0}\ \frac{1}{2}||\beta||^2 \tag{27}$$

$$Y_i\ (\beta.X_i + \beta_0) \geq 1\forall_i \tag{28}$$

For a soft-margin SVM, which allows some misclassification, the optimization problem becomes:

$$min_{\beta,\beta_0,\xi}\frac{1}{2}||\beta||^2\ + C\ \sum_{i=1}^{m}\xi_i \tag{29}$$

where ξ_i slack variables, and C slack is a regularization parameter.

$$f(X) = \sum_{i=1}^{m}\alpha_i\ Y_i K(X_i,\ X) +\ \beta_0 \tag{30}$$

where $K(X_i,\ X)$ is a kernel function.

G. Gradient Boosting

Gradient descent, a step-by-step optimization method, minimizes a cost function by adjusting model parameters to decrease the errors in predictions.

The initial model $F_0\ (X)$ predicts a constant value. For classification, this is usually the mode of the target classes:

$$F_0(X) = \arg\ \max_{\mathrm{yk}}\ \sum_{i=1}^{m}\delta\left(y_i' = yk\right) \tag{31}$$

$r_i^{(t)} = \frac{\vartheta L(Y_i,\ F(X_i))}{\vartheta F(X_i)}$; For each t compute the residuals $r_i^{(t)}$

$$L(Y_i\ F(X_i)) = -Y_i\log(\sigma\ (F(X_i))) -(1 - Y_i)\log(1 - \sigma(F(X_i))) \tag{32}$$

where σ indicates the sigmoid function

$$\sigma(F(X_i)) = \frac{1}{1 + e^{-F(X_i)}} \tag{33}$$

The residuals become:

$$r_i^{(t)} = Y_i -\ \sigma\left(F^{(t-1)}\ (X_i)\right) \tag{34}$$

Calculate model performance metrics such as:

$$Accuracy = \frac{\sum_{i=1}^{n_{test}} 1\ \left(\widehat{y}_{test\ \ i} =\ y_{test,i}\right)}{n_{test}} \tag{35}$$

$$Precision(P) = \frac{\mathrm{TP}}{TP + FP} \tag{36}$$

$$Recall = \frac{TP}{TP + FN} \tag{37}$$

$$F1 = 2 \times \frac{Precision\ \ \times Recall}{Precision + Recall} \tag{38}$$

$$RSME = \sqrt{\frac{1}{n}\ \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2} \tag{39}$$

Compute the MAE:

$$MAE = \frac{1}{n}\ \sum_{i=1}^{n}|y_i - \widehat{y}_i| \tag{40}$$

Compute the R^2:

$$R^2 = 1 - \frac{\sum(y_i -\ \widehat{y}_i)^2}{\sum^2_{(y_i - \widehat{y})}} \tag{41}$$

V. Result and Discussion

The application of the hyperparameter optimization-based grid search algorithm in a crop recommendation system using soil nutrient and historical crop yield predictions significantly enhances model performance by optimizing hyperparameter for accurate crop prediction. The result demonstrates that fine-tuning ML models such as (RF, DT, KNN, RF, SVM, and Gradient Boosting) using grid search leads to increased predictive model accuracy compared to default parameters settings [27]. Train the ML model, using the pre-processed data to predict suitable crops for a given set of environmental and agronomic factors. The data has been divided into two parts: the train set and the test set. 70% of the whole data is chosen for model training, while the rest 30%, is reserved for model testing and validation. We pre-processed the agricultural crop recommendations dataset, soil compositions, and crop names to prepare them for model training. In farming, accurately suggesting the right crops is crucial for achieving the best results and maintaining sustainability. As farmers and experts in agriculture explore models based on data, it becomes clear how important it is to use detailed information, especially about the soil's parameters [26]. This section delves into the findings, evaluating model performance, examining the interpretability of the recommendations, and addressing the limitations of current methodologies.

A. Simulation Set Up

We conducted this experiment through a Google Colab Python environment that facilitates the execution and documentation of Python programming code. Google Colab Notebook was chosen due to its interactive interface, which allows for the seamless integration of Python code execution, visualization, and explanation. The experiment environment was configured with Python version 3.12 as the programming language. All the required libraries and dependencies such as Numpy, Pandas, Scikit-Learn, Matplotlib, XAI, LIME, and SHAP, were pre-installed using pip and conda python packages. This experiment setup ensures a reproducible and well-organized workflow, detailed analysis, and interpretation of the results.

B. Dataset and Data Analysis

We have used a crop recommendation open-source dataset and downloaded it from Kaggle. The crop recommendation dataset encompasses various crop features with 22 crop varieties represented in the dataset; each crop comprises 100 records, encompassing 8 distinct features as shown in Table II.

XAI-based crop recommendation system provides transparency and interpretability in decision-making based on soil

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TABLE II
INPUT PARAMETERS FOR CROP RECOMMENDATION

X_1	Soil pH	X_6	Rainfall
X_2	Nitrogen (N) content	X_7	Humidity
X_3	Phosphorus (P) content	X_8	Crops label
X_4	Potassium (K) content		
X_5	Temperature		

TABLE III
COMPARATIVE ANALYSIS OF MODEL PERFORMANCE

ML Model	Accuracy	Precision	Recall	RMSE	MAE	R^2
LR	96.36	89.74	85.02	96.06	96.64	96.80
DT	98.86	92.08	90.37	98.24	98.04	98.47
KNN	95.90	90.65	91.01	95.36	95.84	95.16
SVM	96.81	91.43	88.72	96.32	96.72	96.12
RF	96.31	87.20	86.46	96.73	96.28	96.13
GB	98.18	93.03	91.16	98.57	98.72	98.85
Grid Search	**99.73**	**95.05**	**94.68**	**99.52**	**99.36**	**99.17**

nutrients and historical crop yields. Traditional AI models often function as closed boxes, making it difficult for farmers to understand why a particular crop is recommended. XAI techniques such as LIME and SHAP, help farmers identify the key factors influencing recommendations [27]. Integrate hyperparameter optimization based grid search algorithm of crop recommendation system by systematically optimizing hyperparameters to improve prediction accuracy using soil nutrient and historical crop yields. The hyperparameter optimization algorithm searches through a predefined set of parameters (learning rate, kernel types in SVM, number of trees in random forest) to find the best combination for the model. Selecting the optimal hyperparameters, the model can better map the relationship between soil nutrients (N, P, K) and past crop yields. Farmers predict the crop based on various input features (N, P, K pH, rainfall, Humidity, and Temp) suppose the nitrogen value is 98 percent and humidity is greater than 86.52%, the P value is 8%, the K value is 51%, rainfall is 49.43%, the pH value is 6.26% and temperature is 26.18% then farmer predict the maize crop. The LIME model provides localized explanation, helping region-specific optimizations. For example, given the soil sample wheat is recommended (90% confidence) because of high Nitrogen (N) and moderate pH level, and Maize is not recommended (10% confidence) due to low potassium (K). This value helps farmers focus on the nutrient adjustments to improve soil quality for other crop selections.

The experiment results are shown in Table III and that shown outperformance the other comparison model across all metrics. Grid search model accuracy is 0.9973, Precision of 0.9505, Recall 0.9468, RMSE values of 0.9952, MAE score of 0.9936, and R^2 score of 0.9917 compared to the other machine learning models. The KNN model achieves the lowest accuracy of 0.9590, RMSE performance of 0.9536, MAE score of 0.9584, and R^2 score of 0.9516.

The grid search model outperforms default configurations by reducing errors and improving generalization across different soil conditions. The best combinations of parameters (N, P, K, pH, Rainfall, Humidity, and Temp.) will be selected based on cross-validation accuracy [24].

Fig. 2 (a) shows the comparison analysis graph of the ML model accuracy of the test data set and Fig. 2 (b) Demonstrates the precision comparison graph of the ML model through the K-fold cross-validation test results. Fig. 2(c) shows the Recall comparison graph of the crop recommendation system that signifies a better model performance. Fig. 2 (d) shows a model RSME graph that evaluates the model performance and preference over accuracy. Additionally, continuous feedback from farmers can be used to refine and improve the model, ensuring its recommendations remain relevant and accurate. Figure 3 shows the feature prediction of the crop probability, which suggests the most suitable crop based on the different climate conditions and soil properties.

Figures 4 & 5 show the explanation of the crop feature importance and the farmer selects the crop accordingly. Transparency is achieved by using interpretable models or incorporating techniques like LIME to break down the smart farming solution process into understandable components.

We have measured how each nutrient affects model accuracy and conducted feature ablation experiments, where one nutrient is removed and accuracy is observed. We have included all the features and the model performance is 99.73% without nitrogen (N) model performance is 91.3%, and without potassium (K) model performance is 86%. Soil nutrients significantly impact crop recommendation accuracy, with N, P, K, pH level rainfall, etc. having the highest influence. A farmer in Punjab, India inputs the following soil data such as high Nitrogen (N) greater than 85% low phosphorus (P) less than 20% moderate potassium (K) greater than 50%, pH level less than 6.5%, soil moisture greater than 80%, and rainfall less than 64% based on the parameters traditional ML model suggest the "maize" with no explanation whenever our XAI based model suggest "maize" with LIME and SHAP insights. The farmers now understand why maize is recommended and can adjust phosphorus levels for better crop productivity.

VI. CONCLUSION AND FUTURE DIRECTION

The agriculture sector plays a crucial role in the country's economy and has experienced significant growth thanks to incorporating advanced thrust technologies like IoT, Cloud computing, Fog/Edge, AI, ML, and Blockchain. However, the agriculture sector still facing challenges like variability in climate, optimal crop prediction, temperature, unawareness about crop rotation, accurate plant health, etc. This study proposed advanced crop recommendation models using machine learning techniques to identify the most suitable crop yield predictions. The study proposes grid search model is both adaptable and scalable, allowing for seamless integration with new datasets and diverse geographical regions. The integration of XAI ensures that recommendations are transparent, trustworthy, and actionable, fostering greater adoption among farmers and agriculture stakeholders. The system can optimize

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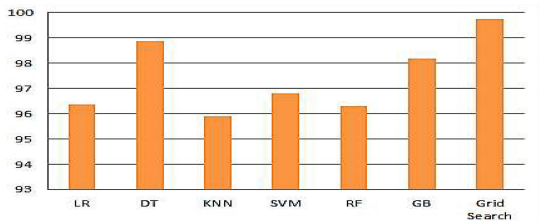
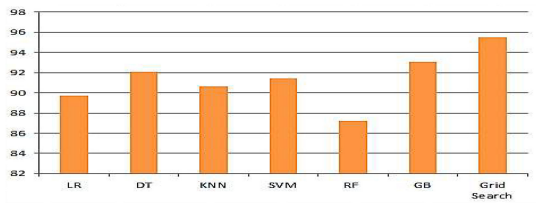
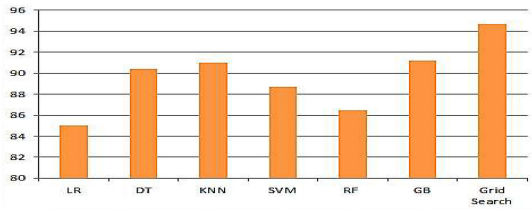


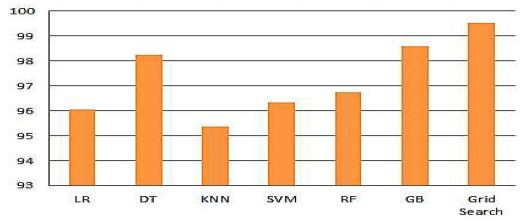
(a) Accuracy comparison graph for crop recommendation



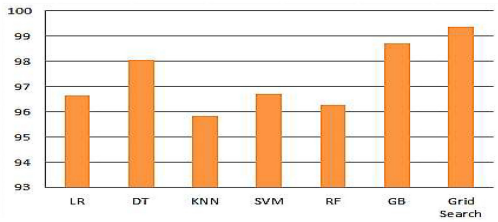
(b) Precision based comparison graph for crop recommendation



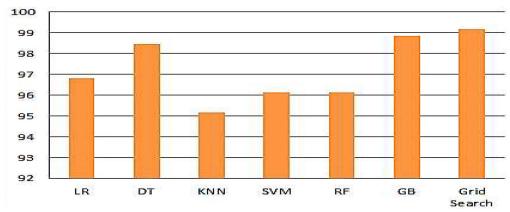
(c) Recall based comparison graph for crop recommendation



(d) RMSE-based comparison graph for crop recommendation



(e) MAE comparison graph for crop recommendation



(f) R^2 comparison graph for crop recommendation

Fig. 2. (a) Accuracy comparison graph for crop recommendation (b) Precision based comparison graph for crop recommendation (c) Recall based comparison graph for crop recommendation (d)RMSE-based comparison graph for crop recommendation (e) MAE comparison graph for crop recommendation (f) R^2 comparison graph for crop recommendation.

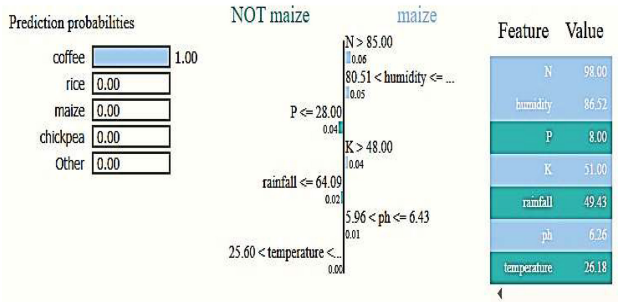


Fig. 3. XAI validates the crop prediction probabilities based on crop features.

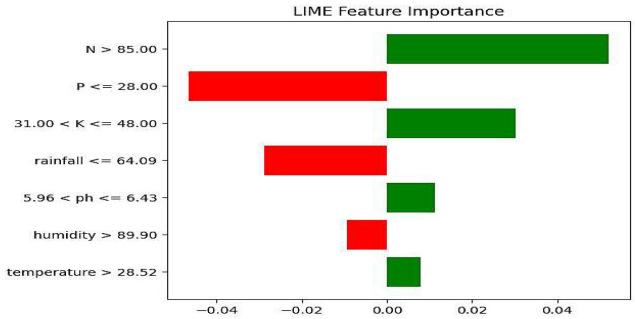


Fig. 5. XAI suggests the feature importance of crops.

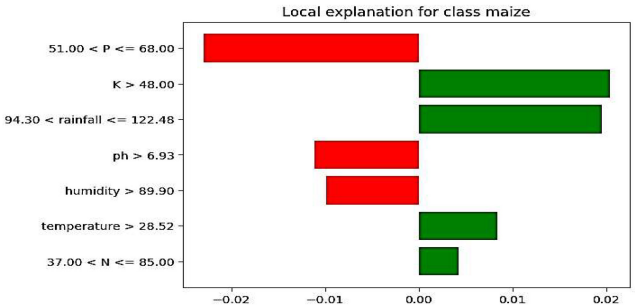


Fig. 4. XAI suggests the feature importance of crops.

resource utilization, enhance crop productivity, and support sustainable farming practices. This framework not only offers high accuracy but is also interpretable, ensuring that the predictions are both reliable and understandable. The model demonstrates significant potential and the system improvement integrates remote sensing data for real-time monitoring of soil health and crop conditions. We will integrate blockchain technology in the framework to enhance transparency, traceability, and efficiency throughout the supply chain. It enables secure and immutable recording of data related to crop production, processing, and distribution, helping ensure food safety and quality. Another future direction of this work is to facilitate real-world applications in remote or low-connectivity regions. The framework can be deployed on mobile or edge devices. Future work should focus on developing intuitive visualizations and natural language explanations that translate technical insights into practical farming decisions.

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